



Natural Language Processing

Orthography

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Natural Language Processing

Orthography

In this lecture:

- 
- ☐ Part 1: **What is Orthography**
 - ☐ Part 2: Arabic Orthography
 - ☐ Part 3: Arabic Spelling and Phonology

What is Orthography

قواعد الكتابة

Most languages evolved as oral languages, then a writing systems are standardized as a way to represent the spoken language.

Orthography is the conventions for writing a language, which include:

- **Spelling rules**, rendering of speech sound (phoneme) into writing (grapheme).
- **Hyphenation**, to join words, e.g., co-operation, de-escalate
- **Capitalization**, first letter in pronouns, abbreviations, sentence beginning ...
- **Word breaks**
- **Emphasis**
- **punctuation**

Graphemes

وحدة خطية / رسوم

- Orthographic units (called graphemes) are the smallest units in a writing system.
- **Types of graphemes** include: alphabetic letters (حروف), syllabic characters (e.g., diacritics تشكيلات), digits (ارقام), punctuations, mathematical symbols, word dividers such as the space.
- Mostly (but not always), a grapheme is a written form or a phoneme (وحدة صوتية)
- Sometimes phoneme represented by a multigraph, e.g., *sh* in English.
- Some graphemes may not represent any sound, e.g., *b* in English debt.
- A grapheme is an abstraction of a collection of **glyphs**, i.e., different glyphs representing the same grapheme. For example (**a**) and (**ɑ**) are different glyphs of the same grapheme.

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In this lecture:

- ☐ Part 1: What is Orthography
- ☐ Part 2: **Arabic Orthography**
- ☐ Part 3: Arabic Spelling and Phonology



Phoenician Alphabet (الأبجدية الفينيقية/الكنعانية)



Letters of the Phoenician alphabet, found in Tel Zayit (Zeitah), 50kmsouthwest Jerusalem.



- The first written system that consists of letters.
- Used across the Mediterranean (Phoenicia, سواحل بلاد الشام) 13th centuries BC.
- uses 22 consonant letters — the vowel sounds were left implicit
- Later it was used in Arabic, Greek, Italic, ...

Phoenician Alphabet (الأبجدية الفينيقية/الكنعانية)

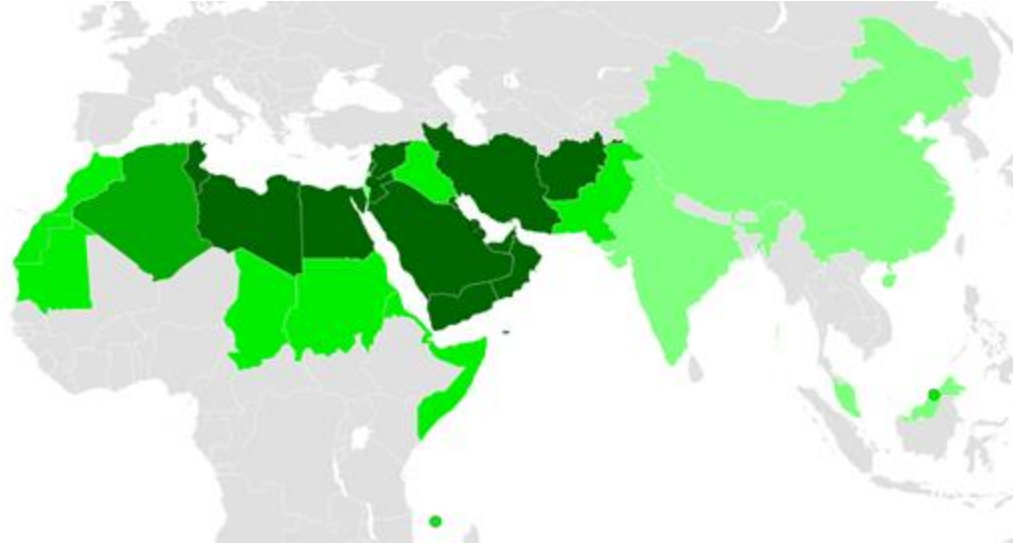
Origin			Letter		Name ^[22]	Meaning	Phoneme	Trans-literation					
Egyptian hieroglyphs	Proto-Sinaitic	Proto-Canaanite	Image	Text					Libyco-Berber	Samaritan	Aramaic	Hebrew	Sy
					'ālep	ox, head of cattle	ʾ [ʔ]	ʾ					
					bēt	house	b [b]	b					
					gīml	throwing stick (or camel) ^[23]	g [g]	g					
					dālet	door (or fish) ^[23]	d [d]	d					
					he	window (or jubilation) ^[23]	h [h]	h					
					wāw	hook	w [w]	w					
					zayīn	weapon (or manacle) ^[23]	z [z]	z					
					ḥēt	courtyard/ wall ^[24] (?)	ḥ [ḥ]	ḥ					
					ṭēt	wheel ^[c]	ṭ [tʰ]	ṭ					
					yod	arm, hand	y [j]	j					
					kāp	palm of a hand	k [k]	k					
					lāmed	goad ^[25]	l [l]	l					

Arabic Script (الخط العربي)

Based on [2]

- Second-most widely used **writing system** (right to left in a cursive style)

- Used for writing **Arabic** and other languages, such as Persian (فارسي), Malay (ملايو), Uyghur (ئۇيغۇرچە), Kurdish (کوردی), Punjabi (پنجابی), Sindhi (سنڌي), Balti (بalti), Balochi (بلوچي), Pashto (پښتو), Lurish (لوري), Urdu (اُردو), Kashmiri (کٲشُر), Rohingya, Somali (صومالي) and Mandinka (مانڊينگا), among others.



Countries where the Arabic script:

- official script
- official script, but other scripts are recognized for national or regional languages
- official alongside other scripts
- official at a sub-national level (China, India) or is a recognized alternative script (Malaysia)

- Several **alphabets** are developed based on the Arabic Script, such as **Arabic**, Persian, Ajami, Aljamiado, and others.

Arabic Alphabet (الأبجدية العربية)

Evolved from the **Nabataean** (النبطية), or maybe from the **Syriac** (السريانية).

The first recorded text in the Arabic Alphabet was written in 512. It is a trilingual dedication in Greek, Syriac and Arabic found at Zabad (الزبداني) in Syria.



	ʾ	b	g	d	h	w	z	ḥ	t	y	k	l	m	n	s	ʾ	p	ṣ	q	r	š	t
Phoenician	𐤀	𐤁	𐤂	𐤃	𐤄	𐤅	𐤆	𐤇	𐤈	𐤉	𐤊	𐤋	𐤌	𐤍	𐤎	𐤏	𐤐	𐤑	𐤒	𐤓	𐤔	𐤕
Aramaic	ܐ	ܒ	ܓ	ܕ	ܗ	ܘ	ܙ	ܠ	ܡ	ܢ	ܫ	ܬ	ܐ	ܦ	ܨ	ܩ	ܪ	ܫ	ܬ	ܐ	ܦ	ܨ
Nabatean	𐤀	𐤁	𐤂	𐤃	𐤄	𐤅	𐤆	𐤇	𐤈	𐤉	𐤊	𐤋	𐤌	𐤍	𐤎	𐤏	𐤐	𐤑	𐤒	𐤓	𐤔	𐤕
Early Arabic	ا	ب	ج	د	هـ	و	ز	ح	ط	ي	ك	ل	م	ن	س	ع	ف	ص	ق	ر	ش	ت
Modern Arabic	ا	ب	ج	د	هـ	و	ز	ح	ط	ي	ك	ل	م	ن	س	ع	ف	ص	ق	ر	ش	ت

Arabic Letters

Based on [1]

Letter Shapes

- No distinction between print and handwriting

- No capitalization

- Right-to-left

- Ambiguous shapes

- Connective letters

ز	د	ا	ن	ب	ك	م	ش	غ	Stand alone
			ز	ب	ك	م	ش	غ	initial
ز	د	ا	ن	ب	ك	م	ش	غ	medial
			ن	ب	ك	م	ش	غ	final

- Disconnective letters (ة ي ر ز د ذ و ؤ ا) cause word internal visual spacing

Arabic Letters

Based on [1]

Letter shaping

كتب = ك ت ب ←
/ka**t**a**b**/
to write

ك ت ب
b t k

كتاب = ك ت ا ب ←
/ki**t**ā**b**/
book

ك ت ا ب
b ā t k

Arabic Diacritics

Based on [1]

- Optionally written
 - Zero-width characters (not stand alone)
 - Used for short vowels
- كَتَبَ / katab/ to write
- Nunation (تنوين) is used for nominal indefinite marker in MSA

كِتَابٌ / kitābun/ a book

Nunation	Vowel
بَ /ban/	بَ /ba/
بُ /bun/	بُ /bu/
بِ /bin/	بِ /bi/

Arabic Diacritics

Based on [1]

- No-vowel marker (sukun)

مَكْتَب /maktab/ office

- Double consonant marker (shadda)

كَتَّب /kattab/ to dictate

- Combinable

بَّ

/bba/

بِ

/bbi/

بُن

/bbun/

No Vowel
بْ
/b/

Double Consonant
بَّبْ
/bb/

Combining letters

Based on [1]

Arab /ʕarab/

عَرَب ← عَرَب = عرب

West /ʁarb/

غَرْب ← غَرْب = غرب

Ligatures

Peace /salām/

س ل ا م ← ~~س ل ل م~~ ← سلام

Special Cases

- (لا) can be written as one character, instead of (ل+ا)
- (ئ) might be written as two letters (ءى)
- Fateh Nunation might be Witten as (کتاباً) or (کتابًا) both are correct

elongation (تطويل)

Based on [1]

- aka kashida
- Visual elongation
- Used for text highlight and justification

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Arabic Fonts

Based on [1]

- Different styles
- High fluidity
- Optional ligatures
- Vertical arrangements

Arabic	Muhammad	algebra
عربي	محمد	الجبر
عربي	محمد	الجبر
عربي	محمد	الجبر
عربي	محمد	الجبر
/ʿarabi/	/muḥammad /	/alǧabr/

Arabic Numerals

Based on [1]

- Decimal system
- Numbers written left-to-right in right-to-left text

استقلت الجزائر في سنة 1962 بعد 132 عاما من الاحتلال الفرنسي



Algeria achieved its independence in 1962 after 132 years of French occupation.

- Three systems of enumeration symbols that vary by region

Western Arabic <i>Tunisia, Morocco, etc.</i>	0	1	2	3	4	5	6	7	8	9
Indo-Arabic <i>Middle East</i>	.	١	٢	٣	٤	٥	٦	٧	٨	٩
Eastern Indo-Arabic <i>Iran, Pakistan, etc.</i>	.	۱	۲	۳	۴	۵	۶	۷	۸	۹

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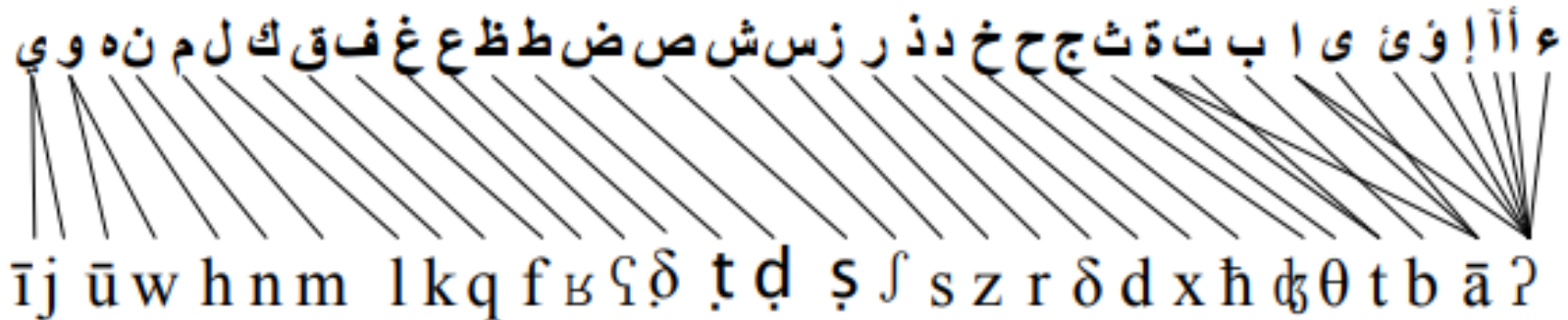
- ☐ Part 1: What is Orthography
- ☐ Part 2: Arabic Orthography
- ☐ Part 3: **Arabic Spelling and Phonology**



MSA Phonology and Spelling

Based on [1]

- Phonological profile of standard Arabic
 - 28 Consonants
 - 3 short vowels, 3 long vowels, 2 diphthongs
- Arabic spelling is mostly phonemic ...
 - Letter-sound correspondence



MSA Phonology and Spelling

Based on [1]

- Arabic spelling is mostly phonemic ...

Except for

- Medial short vowels can only appear as diacritics
- Diacritics are optional in most written text
 - Except in holy scripture
 - Present diacritics mark syntactic/semantic distinctions

كُتِبَ / katab/ to write	كُتِبَ / kutib/ to be written
حُب / ħubb/ love	حَب / ħabb/ seed

- Dual use of (ا ، و ، ي) as consonant and long vowel

MSA Phonology and Spelling

Based on [1]

- Arabic spelling is mostly phonemic ...

Except for

- Morphophonemic characters
 - Feminine marker ة (ta marbuta)
كبير / kabīr/ (big ♂) كبيرة / kabīr a/ (big ♀)
 - Derivation marker
/ʿasa/ (to disobey عصى) (a stick عصا)
- Hamza variants (6 characters (ء آ إؤئ) for one phoneme!)
بهاء بهاء بهاء بهاءه — /baha'/ + 3MascSing (his glory)

MSA Phonology and Spelling

Based on [1]

- Arabic spelling can be ambiguous
 - optional diacritics and dual use of letter
- But how ambiguous? Really?
- Classic example

ths s wht n rbc txt lks lk wth n vwls

this is what an Arabic text looks like with no vowels

- Not exactly true
 - Long vowels are always written
 - Initial vowels are represented by an 'alef'
 - Some final short vowels are represented

ths is wht an Arbc txt lks lik wth no vwls

Will revisit ambiguity in more detail again under morphology discussion

Proper Name Spelling

Based on [1]

- The Qadafi-Schwarzenegger problem
 - Foreign Proper name spelling is often ad hoc
 - Multiplicity of spellings causes increased sparsity

قذافي	→	Gadafi Gaddafi Gaddfi Gadhafi Ghaddafi Kadaffy Qaddafi Qadhafi ...
شوارزنيغر شوارزنغر شوارزنيجر شوارتزنجر	←	Schwarzenegger

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